

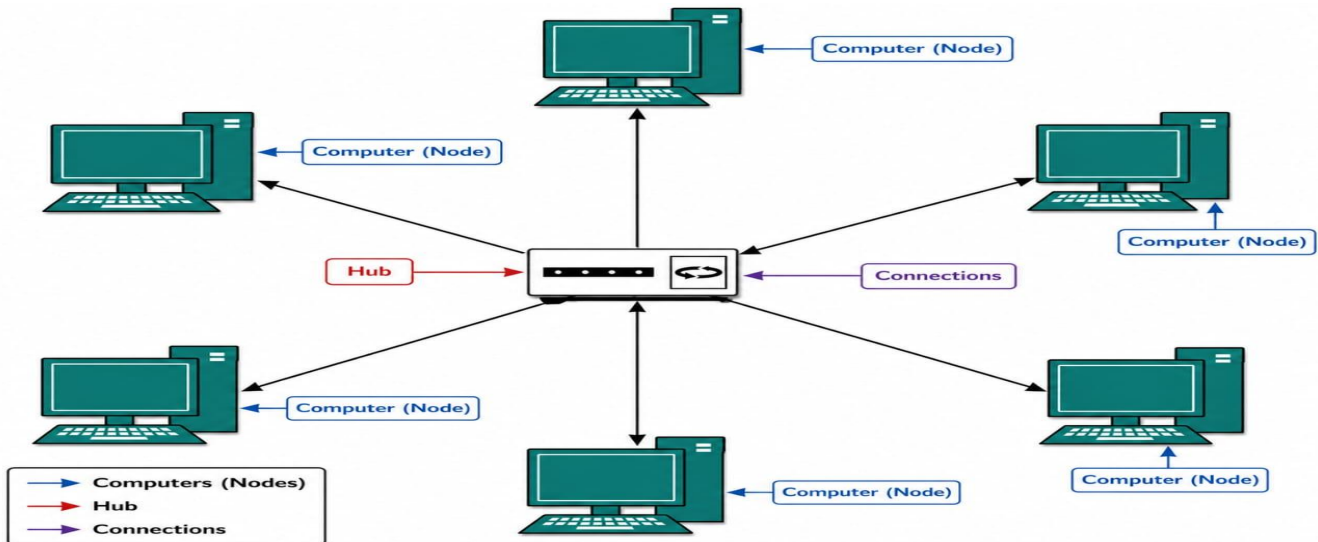
COMPUTER NETWORK

Lesson-7

Class-Seven

INTRODUCTION:

- ➡ A computer network is a set of computers connected together to share resources.
- ➡ Computers connected to a network through different medium are called nodes. The medium might be wired or wireless.
- ➡ Computers can be connected through cables or wirelessly.
- ➡ A network allows sharing of printers, files, modems, and Internet.
- ➡ Internet itself is a computer network.



In short: **A computer network connects devices to communicate and share resources.**

SERVICES OF COMPUTER NETWORK:

A computer network provides services that allow computers to share information, files, hardware and media. Some of them are given as:

1. File Sharing Service:

- Allows computers to share files and documents with each other.

- Users can choose which files/folders to share.

2. Printer Sharing Service:

- Allows computers on a network to use a shared printer.
- A print server can allow multiple computers to use printers.

3. Database Sharing Service:

- Allows network users to store, access and use shared database data.
- Useful for handling large numbers of related records.

4. Media Sharing Service:

- Allows computers to share music, videos, movies and other media.
- Media can be streamed from one computer to another.

COMPONENTS OF COMPUTER NETWORK:

There are five basic components required for communication in a computer network:

1. Data (Message):

The information to be communicated through a network.

Examples: text, pictures, audio and video.

2. Sender:

The device that sends the data/message.

Examples: computer, workstation, telephone handset.

3. Receiver:

The device that receives the data/message from the sender.

Examples: computer, workstation, telephone handset.

4. Transmission Medium:

The path through which data travels from sender to receiver.

Examples: twisted-pair wire, coaxial cable and radio waves.

5. Protocol:

A set of rules that controls data communication.

It provides an agreement between communicating devices.

Examples: HTTP (Hyper Text Transfer Protocol), FTP (File Transfer Protocol) and NCP (Network Control Protocol).

★ Easy way to remember:

Data → Sender → Medium → Receiver → Protocol

ADVANTAGES AND DISADVANTAGES OF COMPUTER NETWORK:

As we know, Computer Network is faster and cheaper with huge facilities but there are some disadvantages of Computer Network. Both Advantages and disadvantages of Computer Network are given as:

Advantages of Computer Network:

1. Provides convenient resource sharing.
2. Makes data communication faster and cheaper.
3. Allows sharing of hardware such as printers, fax and modems.
4. Increases storage capacity.
5. Makes mass data communication faster and cheaper.

Disadvantages of Computer Network:

1. There is a risk of security problems.
2. Initial setup cost can be high.
3. Networks can spread computer viruses and malware.

TYPES OF COMPUTER NETWORK:

Computer networks are mainly classified into three types:

1. LAN – Local Area Network:

A LAN connects computers over a small geographical area, such as a home, school, college or office.

Characteristics:

1. Covers a limited distance.
2. Usually uses physical cables for communication.
3. Generally owned by a single organization.
4. Has a faster data transfer rate than MAN and WAN.



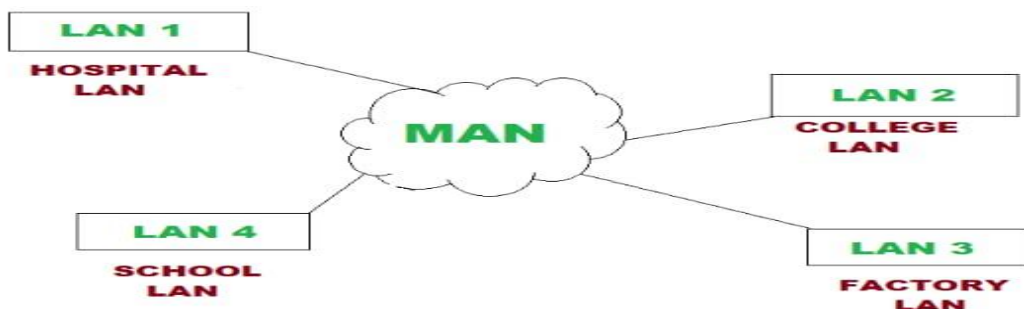
Example: Computer network inside a school.

2. MAN – Metropolitan Area Network

A MAN is larger than LAN but smaller than WAN. It generally covers a city or metropolitan area.

Characteristics:

1. Covers an entire city or state.
2. Uses wired and wireless communication.
3. Usually owned by a single public organization.
4. Data transmission is slower than LAN.



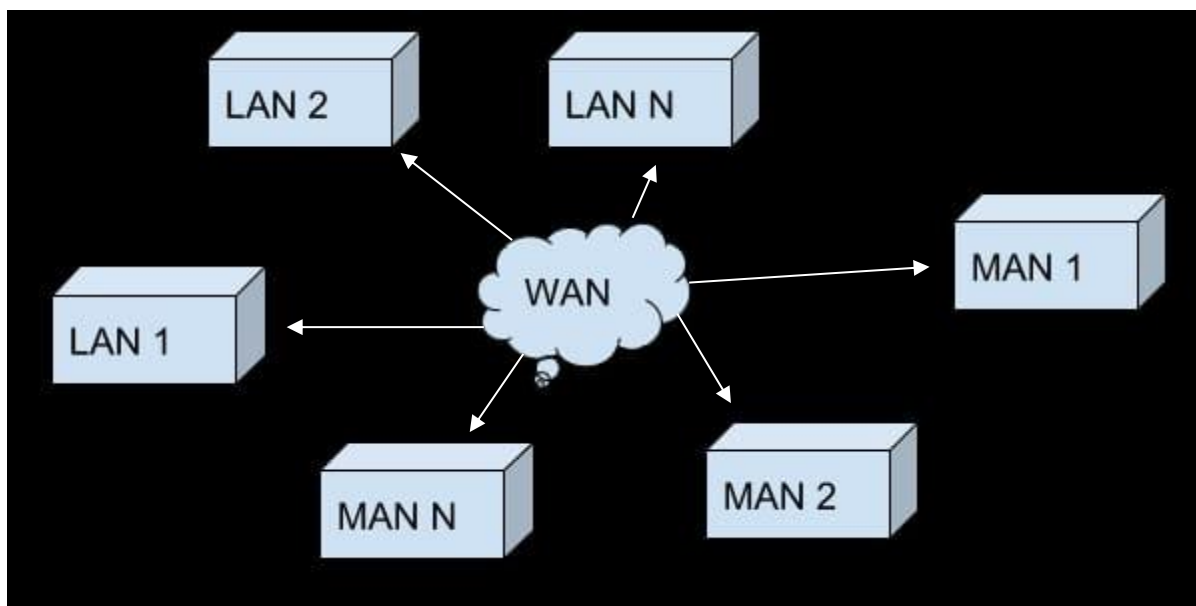
Example: Network connecting offices across a city.

3. WAN – Wide Area Network

A WAN covers a very large geographical area, such as countries or the whole world.

Characteristics:

1. Covers unlimited/global distances.
2. Can use wireless communication for data transmission.
3. May be owned by different public organizations.
4. Data transmission is slower than LAN and MAN.



Example: Internet is an example of WAN.